Terrence Alsup

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Research Interests

Statistical inverse problems, machine learning, Monte Carlo methods, uncertainty quantification, numerical and scientific computing

Education

Sep. 2017 – Jan. 2023 **Ph.D. Mathematics**, *Courant Institute, New York University*, New York, NY
- Advisor: Benjamin Peherstorfer
- Dissertation (defended Dec. 2022):
  Trading off deterministic approximations and sampling in multifidelity Bayesian inference
- Henry MacCracken Fellowship

Jan. 2017 – Aug. 2017 **M.S. Mathematics**, *Georgia Institute of Technology*, Atlanta, GA
- STEM Fellowship

Aug. 2013 – Dec. 2016 **B.S. Applied Mathematics**, *Georgia Institute of Technology*, Atlanta, GA
- Highest honors

Work Experience

Feb. 2022 – Present **Finastra**, *Data Scientist*, Innovation Lab
- Develop statistical and machine learning models to derive insights and aid decision-making within the four primary business units: Payments, Lending, Treasury & Capital Markets, and Universal Banking
- Analyze Canadian mortgage lending data over time for multiple lenders and compare to both peer and market-based benchmarks to identify anomalies

Jun. 2021 – Aug. 2021 **Sandia National Laboratories**, *Computer Science Research Institute intern*
- Bayesian machine learning group with Tommie Catanach focusing on applications of Bayesian optimal experimental design to sensor configuration for seismic monitoring
- Developed a method for estimating the error in expected information gain estimates

Honors & Awards

2022 Vista Hackathon Winner
The Grid: Federated learning for the credit invisible

2020, 2022 SIAM student travel award

2021 BGCE Student Paper Prize Finalist

Publications, Preprints, & Technical Reports

[1] A. Maurais, **T. Alsup**, B. Peherstorfer, and Y. Marzouk. Multi-fidelity covariance estimation in the log-Euclidean geometry. *arXiv:2301.13749*, 2023.

[2] **T. Alsup** and B. Peherstorfer. Context-aware surrogate modeling for balancing approximation and sampling costs in multi-fidelity importance sampling and Bayesian inverse problems. *SIAM/ASA Journal on Uncertainty Quantification*, 2022.

[3] **T. Alsup**, T. Hartland, B. Peherstorfer, and N. Petra. Further analysis of multilevel

Stein variational gradient descent with an application to the Bayesian inference of glacier ice models. *arXiv:2212.03366*, 2022.

[4] **T. Alsup**, L. Venturi, and B. Peherstorfer. Multilevel Stein variational gradient descent with applications to Bayesian inverse problems. In *Mathematical and Scientific Machine Learning (MSML) 2021*, 2021.

[5] **T. Alsup** and T. Catanach. Expected infomation gain estimates and Bayesian optimal experimental design. In J.D. Smith and E. Galvan, editors, *Computer Science Research Institute Summer Proceedings 2021*, pages 269–282, 2021. Technical Report: SAND2022-0653R.

Teaching Experience

2020–2021 Mathematical Statistics, *NYU*, Teaching Assistant (3 semesters)

2019–2020 Introduction to Machine Learning, *NYU*, Class Assistant (2 semesters)

2018 Stochastic Calculus, *NYU*, Teaching Assistant (1 semester)

2014–2015 Differential Equations, *Georgia Tech*, Teaching Assistant (2 semesters)

Conference Presentations

Sep. 2022 SIAM MDS22, *Multilevel Preconditioning for Stein Variational Gradient Descent with Applications to Bayesian Inverse Problems.*

Apr. 2022 SIAM UQ22, *Multilevel Stein Variational Gradient Descent with Applications to Bayesian Inverse Problems.*

Mar. 2022 IMSI, The Mathematics of Soft Matter, *Trading-off deterministic preconditioning and sampling in Bayesian inference.*

Aug. 2021 MSML21, *Multilevel Stein Variational Gradient Descent with Applications to Bayesian Inverse Problems.*

Mar. 2021 SIAM CSE21, *Trading-off Deterministic Preconditioning and Sampling in Bayesian Inference.*

Mar. 2021 SIAM CSE21, *Learning Context-Aware Surrogate Models for Multifidelity Importance Sampling and Bayesian Inverse Problems.*

Apr. 2020 ICERM, Computational Statistics and Data-Driven Modeling, *Learning context-aware surrogate models for multifidelity importance sampling and Bayesian inverse problems.*

Feb. 2020 ICERM, Mathematics of Reduced Order Models, *Learning context-aware surrogate models for multifidelity importance sampling and Bayesian inverse problems.*

Other Projects & Articles

2022 Optimal Transport and its Applications to Fairness, *Published on DZone*

2020 Hedging Momentum with ETFs

2020 Learning Large Graphical Models via Convex Optimization

2019 Parallel Kinetic Monte Carlo, *with Anya Katsevich*

2019 Multi-fidelity Cross-Entropy Estimation of Rare Events, *with Freddy Law*

2016 Image forgery detection in Alipay, *with Ant Financial as a part of RIPS-HK 2016 (report manager)*

2016 Sequential Maximum Likelihood Estimation and Inference for Parameters of Stochastic Differential Equations

Computing skills

Languages: Python (Pandas, Numpy, Scipy, Sci-kit learn, Jax), Matlab, LaTeX, C, Java
HPC: OpenMP, MPI (mpi4py), Slurm, CUDA, Singularity
Software: Git, svn, Gnuplot

Languages

English, *Native*
French, *Intermediate*